

Research Project Proposal

Project Title

BD-DriveIntent: A Bangladesh-Centric Dashcam World Model for Predicting Pedestrian, Rickshaw, and CNG Intent in Mixed Traffic Environments

Alternative Short Title

Explainable Future Traffic Risk Prediction from Bangladesh Dashcam Videos

1. Executive Summary

Road traffic in Bangladesh is highly dynamic, heterogeneous, and difficult to model using conventional autonomous-driving datasets. Unlike structured road environments in many developed countries, Bangladeshi roads often contain mixed traffic participants such as pedestrians, rickshaws, CNG auto-rickshaws, motorcycles, buses, trucks, battery rickshaws, vans, and informal local vehicles. These agents frequently interact in narrow roads, crowded intersections, unmarked lanes, and unpredictable traffic conditions.

This project proposes a Bangladesh-centric dashcam-based world model that predicts the near-future intention and risk behavior of important road participants, especially pedestrians, rickshaws, and CNGs. Instead of only detecting objects in the current frame, the proposed model will observe a short dashcam video clip and predict what may happen in the next few seconds. The model will estimate whether a pedestrian is likely to cross, whether a rickshaw or CNG may cut into the ego vehicle's path, and whether the current situation is safe, cautionary, or high-risk.

The project aims to build a compact but useful Bangladesh dashcam video benchmark with annotated short clips, train a spatio-temporal video model, compare it with strong baselines, and develop an explainable demo system that highlights risky road participants. The final output will include a research paper, dataset documentation, trained models, evaluation results, and a working prototype.

2. Background and Motivation

Autonomous driving and intelligent transportation systems require machines to understand not only what is visible now, but also what is likely to happen next. This is where the idea of a world model becomes important. A world model learns an internal representation of the environment and uses it to predict future states, future object movement, or future risk.

In the context of driving, a world model can be described as:

Past visual observations + road-agent movement + scene context → future intention/risk prediction

Current computer vision models can detect vehicles, pedestrians, traffic signs, and roads. However, detection alone is not enough for safe driving. A dashcam model may detect a pedestrian, but it also needs to predict whether that pedestrian is likely to cross the road. Similarly, detecting a rickshaw is useful, but predicting whether the rickshaw will suddenly move into the vehicle's lane is much more valuable.

Most global autonomous-driving datasets are collected in relatively structured environments such as North America, Europe, Singapore, or other urban regions with better lane discipline and infrastructure. Bangladesh has a significantly different traffic structure. The traffic is denser, more heterogeneous, and more unpredictable. Local vehicles such as rickshaws, CNGs, easy bikes, legunas, and informal small vehicles create unique visual and motion patterns that are not sufficiently represented in many international datasets.

Recent Bangladeshi road-scene datasets such as RSUD20K and BadODD focus mainly on object detection. These datasets are very important, but there is still a research gap in future behavior prediction, intent prediction, near-miss detection, and explainable early-risk forecasting from Bangladesh dashcam videos. This project addresses that gap.

3. Problem Statement

Existing road-scene understanding models often answer the question:

“What objects are present in the scene?”

However, for real-world driving assistance, a more important question is:

“Which object may create risk in the next few seconds?”

In Bangladesh, this question is especially important because road participants often behave unpredictably. Pedestrians may cross without formal crossings, rickshaws may suddenly change direction, motorcycles may overtake from narrow gaps, and CNGs may stop or turn abruptly.

The main problem of this project is therefore:

How can we build a dashcam-based world model that predicts the near-future intention and risk level of pedestrians, rickshaws, and CNGs in Bangladesh's mixed traffic environment?

4. Research Gap

The key research gaps are:

1. Bangladesh-specific future-risk modeling is underexplored.

Existing Bangladesh road datasets are mostly image-based and object-detection oriented. There is limited work on video-based future risk or intention prediction in Bangladeshi traffic.

2. Object detection does not solve early-risk prediction.

A model may detect a pedestrian or CNG accurately, but it may not know whether that object will become dangerous in the next 1–3 seconds.

3. Global datasets may not generalize well to Bangladesh.

Traffic behavior, road structure, vehicle classes, and movement patterns differ significantly across countries. A model trained only on foreign datasets may fail to understand local road behavior.

4. Near-miss events are more practical than accident-only modeling.

Actual accident data is rare, sensitive, and difficult to collect. Near-miss and risky-intent events are more common and more useful for early warning systems.

5. Explainability is often missing.

Safety-critical systems should not only say “high risk”; they should also indicate which object is causing the risk and why.

5. Research Objectives

The main objectives of this project are:

1. To collect and curate a small but meaningful Bangladesh dashcam video dataset for mixed-traffic risk prediction.
2. To annotate short clips with labels such as:
 3. Safe
 4. Caution
 5. High Risk
 6. Near-Miss
7. To create object-level intent labels for key road participants:
 8. Pedestrian crossing intention
 9. Rickshaw cut-in intention

10. CNG sudden-turn or sudden-stop intention
 11. To develop a video-based world model that predicts near-future risk from past dashcam observations.
 12. To compare the proposed model with baseline methods such as CNN-LSTM, 3D CNN, YOLO + LSTM/GRU, and Video Transformer-based models.
 13. To develop an explainable visualization system that highlights risky objects and shows predicted risk level.
 14. To prepare a research paper suitable for submission to a computer vision, intelligent transportation, or applied AI venue.
-

6. Research Questions

This project will investigate the following research questions:

1. Can a dashcam video model predict future traffic risk in Bangladesh's mixed road environment using only past visual observations?
 2. Does Bangladesh-specific fine-tuning improve risk prediction compared to models trained only on global driving datasets?
 3. Which road participants contribute most to early traffic risk in Bangladesh: pedestrians, rickshaws, CNGs, motorcycles, buses, or other vehicles?
 4. Can object-level motion history improve future-risk prediction compared to frame-level video classification?
 5. Can an explainable model localize the risky road participant responsible for a high-risk prediction?
-

7. Proposed System Overview

The proposed system will take a short dashcam video clip as input and generate three types of output.

Input

A short video clip of approximately 3–5 seconds from a vehicle-mounted dashcam.

Output

1. **Scene-level risk prediction**

2. Safe

3. Caution

4. High Risk

5. Near-Miss

6. Object-level intent prediction

7. Pedestrian likely to cross

8. Rickshaw likely to cut in

9. CNG likely to stop or turn suddenly

10. Motorcycle/bus/other vehicle likely to create risk

11. Explainability output

12. Bounding box or heatmap around the risky object

13. Short explanation such as:

- "Pedestrian moving toward road"
- "Rickshaw entering vehicle path"
- "CNG slowing down near ego path"
- "Motorcycle overtaking from narrow gap"

8. Proposed Methodology

The project will be divided into six major stages.

Stage 1: Literature Review

The literature review will focus on:

1. Dashcam-based traffic accident detection
2. Traffic anomaly detection
3. Pedestrian intention prediction
4. Object tracking and trajectory prediction
5. Video-based world models
6. Explainable AI for safety-critical computer vision
7. Bangladesh road-scene understanding datasets

Important reference directions include:

- BDD100K for large-scale driving video understanding

- nuScenes and Waymo Open Dataset for autonomous driving benchmarks
 - PIE dataset for pedestrian intention estimation
 - DoTA dataset for traffic anomaly detection in driving videos
 - RSUD20K and BadODD for Bangladesh road object detection
-

Stage 2: Dataset Collection and Curation

The dataset will be collected from Bangladesh road environments using dashcam or smartphone-mounted camera footage.

Target Collection Plan

For a capstone-level project, the recommended raw footage target is:

- **Minimum:** 20–50 hours
- **Good target:** 100–200 hours
- **Ambitious target:** 300–500 hours

The footage should cover:

- Urban roads
- Narrow roads
- Highways
- Intersections
- Market areas
- Daytime scenes
- Nighttime scenes
- Rainy or low-light scenes, if possible

Important Road Participants

The dataset should include:

- Pedestrian
 - Rickshaw
 - CNG auto-rickshaw
 - Motorcycle
 - Bus
 - Truck
 - Private car
 - Van
 - Easy bike or battery rickshaw
 - Bicycle
 - Leguna or local transport
 - Roadside obstacle
-

Stage 3: Clip Extraction

Instead of training on full long videos, the project will extract short clips.

Clip Length

- 3 seconds
- 5 seconds
- optionally 7 seconds for complex situations

Clip Categories

1. Safe normal driving
2. Pedestrian crossing or preparing to cross
3. Rickshaw entering or cutting into path
4. CNG sudden stop or turn
5. Motorcycle close overtake
6. Bus or truck sudden movement
7. Near-miss or very close interaction
8. Rain/night/low visibility risk

This clip-based approach reduces storage, annotation cost, and training time.

Stage 4: Annotation Strategy

The annotation will have two levels: scene-level and object-level.

A. Scene-Level Labels

Each clip will be labeled as:

1. **Safe**
No immediate risk in the next few seconds.
2. **Caution**
A road participant may create potential risk, but not immediately dangerous.
3. **High Risk**
The situation is likely to require braking, slowing, or steering adjustment.
4. **Near-Miss**
A collision is narrowly avoided or there is a very close interaction.

B. Object-Level Labels

For key road participants, bounding boxes and intent labels may be added.

Example labels:

- Pedestrian crossing: yes/no
- Rickshaw cut-in: yes/no
- CNG sudden-turn: yes/no
- CNG sudden-stop: yes/no
- Motorcycle risky overtake: yes/no
- Risk-causing object ID

C. Temporal Labels

For risky clips, the approximate time of risk onset should be marked.

Example:

- Risk begins at 2.1 seconds
- Near-miss occurs at 4.3 seconds

This will allow evaluation of early prediction performance.

9. Model Architecture

The proposed model can be designed in a modular way.

Module 1: Object Detection

A pretrained object detector will be used to detect road participants.

Possible models:

- YOLOv8
- YOLOv11
- RT-DETR
- Faster R-CNN
- DETR-based models

For Bangladesh-specific classes, the detector can be fine-tuned using RSUD20K, BadODD, or the collected local dataset.

Module 2: Object Tracking

Detected objects will be tracked across frames.

Possible tracking methods:

- ByteTrack

- DeepSORT
- OC-SORT
- BoT-SORT

Tracking will help estimate object motion, direction, and speed.

Module 3: Spatio-Temporal Feature Encoder

The video sequence will be represented using one of the following:

- CNN + LSTM
- CNN + GRU
- 3D CNN
- TimeSformer
- VideoMAE
- SlowFast Network
- ViViT-style video transformer

Module 4: World Model / Future Prediction Head

This is the main research component.

The model will learn to predict future risk based on past observations.

Possible outputs:

1. Future trajectory of selected objects
2. Future bounding box displacement
3. Object-level intent probability
4. Scene-level risk probability
5. Future risk heatmap

A simplified formulation:

Input: frames from time $t-3$ to t

Prediction: risk and object behavior from time t to $t+3$

Module 5: Explainability Head

The model should identify which object caused the risk prediction.

Possible methods:

- Attention map visualization
- Grad-CAM for video model
- Object-level risk score
- Bounding-box risk ranking

Example output:

Object	Class	Risk Score	Explanation
Object 3	Pedestrian	0.87	Moving toward road
Object 5	Rickshaw	0.76	Entering ego path
Object 8	CNG	0.69	Sudden stop likely

10. Baseline Models

To prove the effectiveness of the proposed model, it will be compared with multiple baselines.

Baseline 1: Frame-Based CNN Classifier

Each frame is classified independently.

Expected limitation: does not understand motion well.

Baseline 2: CNN + LSTM

CNN extracts frame features, LSTM models temporal changes.

Baseline 3: YOLO + LSTM/GRU

Object detections and positions are passed to a temporal model.

Baseline 4: 3D CNN

A 3D convolutional network directly learns spatio-temporal video features.

Baseline 5: Video Transformer

Transformer-based video model for temporal attention and long-range interaction.

Proposed Model

Object-aware world model with future-intent and risk-prediction heads.

11. Evaluation Metrics

The project will use both classification and forecasting metrics.

Scene-Level Risk Classification

- Accuracy
- Precision
- Recall
- F1-score
- Macro-F1
- ROC-AUC
- Confusion matrix

Macro-F1 is important because risky/near-miss events may be fewer than safe events.

Object-Level Intent Prediction

- Precision
- Recall
- F1-score
- Average Precision
- Object-level risk localization accuracy

Future Trajectory / Bounding Box Prediction

- Average Displacement Error
- Final Displacement Error
- Intersection-over-Union for future box prediction

Early Risk Prediction

- Time-to-event prediction
- Early warning accuracy
- Risk prediction before near-miss event

Example:

The model should predict high risk at least 1–2 seconds before the actual near-miss moment.

12. Expected Novelty

The novelty of this project will come from four main areas.

1. Bangladesh-Centric Future Risk Dataset

Most Bangladeshi road datasets focus on object detection. This project will create a video-based risk and intent dataset for Bangladesh road scenes.

2. Mixed-Traffic Intent Prediction

The project will focus on local traffic participants such as pedestrians, rickshaws, and CNGs, which are highly important in Bangladesh but underrepresented in global autonomous-driving datasets.

3. World Model for Early Risk Forecasting

The model will not only classify the present scene. It will predict future risk based on past observations.

4. Explainable Risk Localization

The system will identify which road participant is creating the risk and provide visual explanation.

13. Dataset Size Plan

A realistic dataset plan is given below.

Version 1: Minimum Viable Dataset

- Raw footage: 20–50 hours
- Selected clips: 2,000–5,000
- Clip length: 3–5 seconds
- Labels: Safe, Caution, High Risk, Near-Miss
- Suitable for prototype/demo

Version 2: Good Capstone Dataset

- Raw footage: 100–200 hours
- Selected clips: 10,000–30,000
- Clip length: 3–5 seconds
- Scene-level labels
- Object-level labels for risky clips
- Suitable for conference/workshop submission

Version 3: Strong Research Dataset

- Raw footage: 300–500 hours
- Selected clips: 30,000–70,000
- Detailed object-level labels
- Temporal risk onset labels
- Suitable for journal extension

For this project, Version 2 is recommended as the main target.

14. Storage and Computing Requirement

Storage Requirement

For 100–200 hours of selected dashcam footage and processed clips:

- Local active storage: 1–2 TB SSD/NVMe
- Backup storage: 4–8 TB HDD
- Cloud storage: 3–6 TB

A 5–6 TB cloud storage plan is recommended for safe backup and collaboration.

GPU Requirement

Minimum:

- NVIDIA RTX 3060/4060 with 12 GB VRAM for small experiments

Recommended:

- NVIDIA RTX 3090/4090 with 24 GB VRAM
- University GPU server
- Cloud GPU such as A100/T4/L4 depending on budget

Training can be optimized by:

- Using pretrained models
- Freezing backbone layers
- Training on selected clips
- Using mixed precision
- Using lower resolution such as 640×360 or 720p
- Sampling fewer frames per clip

15. Ethical and Privacy Considerations

Since dashcam footage may contain people, vehicles, license plates, and public-road scenes, privacy and ethical handling are important.

The project will follow these steps:

1. Blur faces and license plates before public release.
2. Avoid recording private places or sensitive locations.
3. Use footage only for academic research.
4. Store raw footage securely.
5. Release only anonymized clips if the dataset is made public.
6. Avoid personally identifiable information.

7. Use consent where applicable for any privately collected footage.

16. Implementation Plan

Step 1: Literature Review

Review papers and datasets related to:

- Traffic anomaly detection
- Pedestrian intention prediction
- Autonomous driving world models
- Bangladesh road-scene understanding
- Video transformers and temporal models

Step 2: Data Collection

Collect local dashcam footage from different Bangladesh road environments.

Step 3: Clip Selection

Extract short clips containing safe, caution, high-risk, and near-miss scenarios.

Step 4: Annotation

Create annotation guidelines and label the selected clips.

Step 5: Object Detection Fine-Tuning

Fine-tune YOLO/RT-DETR on Bangladesh-specific road object classes.

Step 6: Tracking

Use object tracking to obtain object motion trajectories.

Step 7: Baseline Training

Train baseline models:

- CNN
- CNN-LSTM
- YOLO + LSTM
- 3D CNN
- Video Transformer

Step 8: Proposed World Model

Train the proposed object-aware future-risk prediction model.

Step 9: Explainability

Add risky-object visualization using attention, Grad-CAM, or object-level risk scoring.

Step 10: Evaluation and Paper Writing

Evaluate all models, prepare result tables, write analysis, and prepare the final paper.

17. Expected Results

The expected results are:

1. A Bangladesh dashcam video dataset for future-risk prediction.
 2. A trained world model that predicts early traffic risk.
 3. Better performance than frame-only and simple temporal baselines.
 4. Explainable output showing the risky object.
 5. A working demo where a dashcam clip is uploaded and the system outputs risk level and risky-object localization.
 6. A research paper suitable for workshop, conference, or journal submission.
-

18. Possible Demo System

The final demo can be a web-based or desktop application.

Demo Input

User uploads a 3–5 second dashcam video.

Demo Output

- Risk level: Safe / Caution / High Risk / Near-Miss
- Risk score
- Risky object bounding box
- Predicted intent
- Short explanation

Example:

Input: Dashcam clip showing a rickshaw entering the vehicle path

Output:

Risk Level: High Risk

Risky Object: Rickshaw

Explanation: Rickshaw is moving laterally toward the ego vehicle's path.

19. Timeline

Month 1: Literature Review and Problem Finalization

- Read related papers
- Finalize scope
- Prepare annotation guideline
- Select public datasets for pretraining/baseline

Month 2: Data Collection and Clip Extraction

- Collect local footage
- Extract short clips
- Organize data folders
- Start initial annotation

Month 3: Dataset Annotation and Baseline Models

- Complete first round annotation
- Train object detector
- Train frame-based and CNN-LSTM baselines

Month 4: Proposed Model Development

- Add object tracking
- Train object-aware temporal model
- Add future-risk prediction head

Month 5: Experiments and Explainability

- Run ablation studies
- Compare with baselines
- Add heatmap/object-level explanation
- Improve model performance

Month 6: Final Paper, Demo, and Presentation

- Prepare final report
 - Prepare research paper
 - Prepare demo video
 - Prepare GitHub repository
 - Prepare final presentation
-

20. Team Role Distribution

A 3-member team is ideal.

Member 1: Machine Learning and Model Lead

Responsibilities:

- Model architecture
- Training pipeline
- Baseline experiments
- Evaluation metrics
- Result analysis

Member 2: Dataset and Annotation Lead

Responsibilities:

- Dashcam footage collection
- Clip extraction
- Annotation guideline
- Data cleaning
- Dataset documentation

Member 3: System, Demo, and Paper Lead

Responsibilities:

- Demo application
- Visualization
- Literature review support
- Paper writing
- GitHub documentation

If a 4th member is added, they can handle backend/data pipeline and cloud storage management.

21. Tools and Technologies

Programming

- Python
- PyTorch
- OpenCV
- NumPy
- Pandas

Computer Vision Models

- YOLOv8 / YOLOv11
- RT-DETR
- VideoMAE
- TimeSformer
- SlowFast
- CNN-LSTM / GRU

Annotation Tools

- CVAT
- Roboflow
- Label Studio
- VIA Annotation Tool

Tracking Tools

- ByteTrack
- DeepSORT
- BoT-SORT
- OC-SORT

Experiment Tracking

- Weights & Biases
- TensorBoard
- MLflow

Deployment

- Streamlit
- Flask
- FastAPI
- Gradio

22. Risk Management

Risk 1: Dataset Collection Becomes Difficult

Mitigation:

- Use public datasets for pretraining
- Use YouTube/dashcam videos only if license and ethics are handled properly
- Reduce scope to 20–50 hours for prototype

Risk 2: Annotation Takes Too Much Time

Mitigation:

- Use semi-automatic annotation
- Detect objects using pretrained YOLO
- Manually verify only important clips
- Focus object-level annotation only on risky clips

Risk 3: GPU Limitation

Mitigation:

- Use pretrained models
- Use lower resolution
- Freeze model backbone
- Use frame sampling
- Use cloud GPU for final experiments only

Risk 4: Model Performance Is Low

Mitigation:

- Start with simple baselines
- Use object tracking features
- Balance dataset classes
- Use data augmentation
- Use transfer learning from public datasets

Risk 5: Privacy Concern

Mitigation:

- Blur faces and license plates
- Avoid publishing raw identifiable footage
- Release only anonymized samples

23. Expected Contributions

The project is expected to contribute:

1. A Bangladesh-centric dashcam video dataset for early traffic risk prediction.
2. A benchmark for pedestrian, rickshaw, and CNG intent prediction.
3. A world-model-based architecture for future traffic risk forecasting.
4. Explainable risky-object localization.
5. Experimental comparison with strong baselines.

6. A practical demo system for intelligent driver assistance.

24. Potential Publication Direction

The project can be prepared for:

1. Computer vision workshops
2. Intelligent transportation conferences
3. Applied AI journals
4. Autonomous driving workshops
5. Q1/Q2 journal extension after improving dataset size and experiments

Possible venue categories:

- CVPR/ICCV/ECCV Workshops
 - NeurIPS/ICML Workshops related to world models, embodied AI, or autonomous driving
 - IEEE Intelligent Transportation Systems Conference
 - IEEE Intelligent Vehicles Symposium
 - Applied AI and intelligent transportation journals
-

25. Conclusion

This project proposes a Bangladesh-centric dashcam world model for predicting future traffic risk and road-agent intent in mixed traffic environments. The project is motivated by the unique traffic conditions of Bangladesh, where pedestrians, rickshaws, CNGs, buses, motorcycles, and informal vehicles interact in complex and unpredictable ways.

Unlike standard object detection projects, this research focuses on future prediction: identifying what may happen next and which object may create risk. The expected output is not only a model but also a dataset, benchmark, explainable demo, and research paper.

The project is ambitious but feasible if the scope is controlled properly. Instead of collecting thousands of hours of footage, the recommended approach is to collect 100–200 hours of raw local dashcam footage and curate 10,000–30,000 short clips for annotation and training. This makes the project suitable for a final-year capstone while still maintaining strong research value for foreign scholarship and PhD applications in computer vision.

References to Study

1. Yu et al., “BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning.”
2. Caesar et al., “nuScenes: A Multimodal Dataset for Autonomous Driving.”
3. Sun et al., “Scalability in Perception for Autonomous Driving: Waymo Open Dataset.”

4. Rasouli et al., "PIE: A Large-Scale Dataset and Models for Pedestrian Intention Estimation and Trajectory Prediction."
5. Yao et al., "When, Where, and What? A New Dataset for Anomaly Detection in Driving Videos."
6. Zunair et al., "RSUD20K: A Dataset for Road Scene Understanding in Autonomous Driving."
7. Baig et al., "BadODD: Bangladeshi Autonomous Driving Object Detection Dataset."
8. Recent literature on world models for autonomous driving, future scene prediction, and video-based risk forecasting.